

Eddie Tang

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Hardware-focused electrical engineering student with 3+ years of hands-on PCB and embedded firmware experience for 3V-24V avionics. Proven leader (team lead, VP) who drives cross-functional projects from prototype to product with strong bench validation and simulation skills.

EDUCATION

University of Illinois Urbana-Champaign

Urbana, IL

BS in Electrical Engineering | GPA: **4.0** | James Scholar Honors

Aug. 2025 – May 2028

Minor: Computer Science & Semiconductor Engineering

Montgomery High School (MHS)

Skillman, NJ

High School Diploma | GPA: **3.92** | SAT: 1560 | 12 AP exams (all scores 5)

Sept. 2021 – Jun. 2025

RELEVANT COURSEWORK

Analog Signal Processing, Intro to Computing, Differential Equations, Multivariable Calculus, Quantum Physics

PROJECTS

CAM-MK3 | Illinois Space Society - Spaceshot Avionics | KiCAD, C++, ESP32-IDF Oct. 2025 – Present

- Spearheaded 3rd major revision of a compact live-video transmitter PCB (60×40 mm) for rocket avionics; co-designed schematic, layout, and power architecture to meet size, thermal, and EMI constraints.
- Built and debugged multiple prototypes (reflow + hand-solder) and validated signal & power integrity with oscilloscope, digital scope probes, and bench power supplies.
- Authored I²C and SPI firmware drivers for TVP5151 analog decoder and ESP32-P4 camera interface; validated NTSC→digital pipeline producing 720×480 color frames (YUV422/YUV420).
- Integrated on-device H.264 encoding on ESP32-P4 and implemented near-complete UVC USB video streaming to host - enabling low-latency telemetry for flight tests.
- Implemented cross-board I²C control and 433 MHz-band communication for board status, camera toggling, and live-video feed; board packaged for rocket avionics bay and targeted for upcoming flight test.

Other PCBs | Illinois Space Society - Spaceshot Avionics | KiCAD, C++

Oct. 2025 – Present

- Designed GATOR power-distribution and ground-station PCBs to manage 24V systems and drive Yagi antenna servos; improved tracker reliability and maintainability through modular connector and test-point placement.

TARC Avionics V1–V3 | EasyEDA, C++, Python

Jun. 2022 – Jun. 2025

- Designed, assembled, and programmed custom PCBs and a 6-DOF AHRS using BMI088 IMU and MS5607 barometer for active flight control; achieved a reproducible 80% mission success across iterative flights.
- Developed quaternion-based sensor fusion & algorithm to integrate acceleration for accurate altitude estimation.
- Built Python software-in-the-loop simulations to validate control logic using historical flight telemetry.
- Ranked top 50 nationally at TARC Nationals (1,000 teams); qualified twice for Nationals.

EXPERIENCE & LEADERSHIP

Electric Sensors Project Lead | Illini Solar Car

Jan. 2026 – Present

- Lead an 8-person team to design a Tire Pressure Monitoring System (TPMS) for ~80 psi solar car wheels.
- Researched implementation of low-power operations, system-design, and protocols for BLE, IMUs, and others.
- Coordinated schedules and deliverables across hardware, firmware, and test subteams to meet project milestones.

Research Assistant (Paid), Mechanical Engineering | Rowan University

Summer 2025

- Investigated effects of point defects in thermoelectric materials on lattice thermal conductivity using MD.
- Employed Green-Kubo method to calculate thermal conductivities from heat flux autocorrelation functions.
- Reduced simulation-to-data turnover times by creating Slurm workflow with Matplotlib, LAMMPS, and ASE.
- First author on in-review manuscript investigating point defect effects on thermal conductivity in Si and Ge.

Laboratory Learning Program, Mechanical & Aerospace Department | Princeton University

Summer 2024

- Performed DFT calculations on WO₃ surfaces for plasma-assisted NH₃ synthesis; analyzed results with NumPy.
- Co-authored a peer-reviewed [publication](#) in ACS Energy Letters (Impact Factor: 19.3).

Vice President, Aerospace Club & Founder, Rocketry Team | MHS

Sept. 2023 – Jun. 2025

- Led weekly meetings with 30+ students to engineer rockets; raised \$15K+ through STEM fundraiser initiatives.
- Increased member attendance and participation by 10% by fostering a close-knit, community-first team culture.

SKILLS

Hardware & Tools: PCB schematic/layout (KiCAD, Eagle), reflow & prototype assembly, bench testing equipment

Languages & Firmware: C++, Python, C, Bash, Dart, Go, Java

Software & Frameworks: MCU IDEs (PlatformIO, MCUxpresso), VASP, LAMMPS, ASE, Slurm, Git, Jupyter, OnShape, NX Siemens, Excel

Other: HAM Radio Technician (KE2CNQ); Fluent in English and Chinese